

Goal. Detect the large symbols on piping & instrumentation diagrams (P&IDs): **tanks, pumps and other major equipment** (exchangers, vessels). Line tracing is out of scope. Small symbols (valves, instruments, fittings) are moved to a separate second model.

Work done

1. **Two-model class split.** The *large* model predicts tank, pump and equipment. A PID2Graph `general` box counts as equipment if $\sqrt{\text{area}} \geq 25$ px at 2048 px; it is not equipment if it comes from the Dataset-PID subset. The split assigns every box to exactly one model: 4,558 large and 59,718 small out of 64,276.
2. **Evaluation protocol** (T1): fixed 12-sheet real test set (OPEN100); per-class and class-agnostic AP50 (300-detection limit); recall by object size; 95% bootstrap CIs over the 12 sheets (500 resamples); fp32 scoring. A scoring bug (100-detection cap) was found in code review and fixed; all numbers below come from re-scoring after the fix.
3. **Training changes for large objects:**
 - Random zoom crops up to 4×, with model selection on a zoomed validation set (T2).
 - **T5:** zoom crops cut from 4096 px renders, plus stroke thinning. Measured line width: real sheets 0.70 px; synthetic 1.9 px at 2× zoom and 3.8 px at 4×. Thinning brings synthetic lines to 0.71–0.92 px.
 - **T6:** copy-paste of tank/pump/equipment instances at real-sheet sizes. Real tanks are 60–276 px, but synthetic ones only 36–72 px.
 - **T7:** loader for hand-labelled real sheets, with a guard against test-set leakage. It is ready but not yet used.
4. **Engineering:**
 - All GPU training runs on the NRP Nautilus cluster: parallel jobs on shared storage, job setup cut from about 9 min to under 1 min, and regression tests.
 - Every change was reviewed (5 code reviews) before results were reported.

Model and data

- **Detector:** torchvision Faster R-CNN, ResNet-50-FPN v2, COCO-pretrained. Input resized to 2048 px on the long side.
- **Training:** SGD (lr 0.02, momentum 0.9), batch size 2, mixed precision, 12 epochs.
- **Data:** PID2Graph [1]: 635 synthetic training sheets, 115 synthetic validation sheets; test = 12 real OPEN100 sheets (35 tanks, 16 pumps, 31 equipment boxes).
- **Runs compared:** baseline (one 5-class model on all symbols); large reference (zoom only, 3 seeds); single-seed screens of T5 and T6; **T5+T6 with 3 seeds** (seeds 1–2 on RTX A4000 GPUs, seed 0 on a different card type).

Results

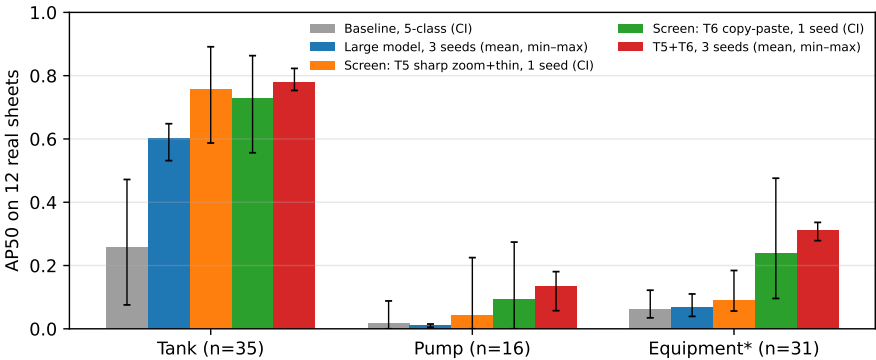


Figure 1: AP50 per class on the 12 real test sheets. Error bars show 95% bootstrap CIs over sheets for single runs, and the min–max range over 3 seeds for the large reference and T5+T6. *The baseline’s equipment bar is its `general` class, a broader definition.

Run (test, 12 real sheets)	Tank AP50	Pump AP50	Equip. AP50	mAP50	Tanks found	Equip. found
Baseline (5-class)	0.26	0.02	0.06*	—	10/35	—
Large ref (3 seeds, mean)	0.60	0.01	0.07	0.23	24–27/35	9–11/31
T5 sharp zoom + thinning	0.76	0.04	0.09	0.30	27/35	12/31
T6 copy-paste	0.73	0.09	0.24	0.35	29/35	21/31
T5 + T6 (3 seeds, mean)	0.78	0.14	0.31	0.41	29/35	22–26/31

Detections are counted at score 0.5 and IoU 0.5. The baseline’s mAP50 is left out because it averages 5 classes and is not comparable.

Findings.

- **Tanks:** focusing the model on large objects more than doubled tank AP50 on real sheets, from 0.26 [0.08, 0.47] to 0.60 (3-seed range 0.53–0.65).
- **Equipment:** copy-paste at realistic sizes (T6) is the main driver. T5+T6 reaches 0.31 (3-seed range 0.28–0.34), against 0.07 for the reference, and finds 22–26 of 31 equipment boxes.
- **Combined setup (3 seeds):** T5+T6 is the best configuration: tank 0.78 (range 0.75–0.82), equipment 0.31, mAP50 0.41 (0.40–0.42); all 3 seeds find 29 of 35 tanks. The seed spread is small, so the gain over the reference is not a lucky seed.
- **Pumps:** still largely unsolved, at 0.14 (seed range 0.06–0.18; 0–2 of 16 found). Real pumps are 33–85 px, but synthetic pumps are only 25–33 px.

Limitations. Small test set (12 sheets, one source; differences under about 0.1 AP are noise); T5-only and T6-only are single-seed screens; copy-paste raises false alarms (121–181 extra equipment boxes); pipe widths still differ from real sheets (0.69 px Wasserstein); no real sheets used in training yet.

Next steps. (1) adopt T5+T6 as the large-model default; (2) label 20–40 real sheets for training (T7 loader); (3) address pump scale and false alarms; (4) train the small-symbol model and define how to merge the two models’ outputs.

Tests on outside drawings

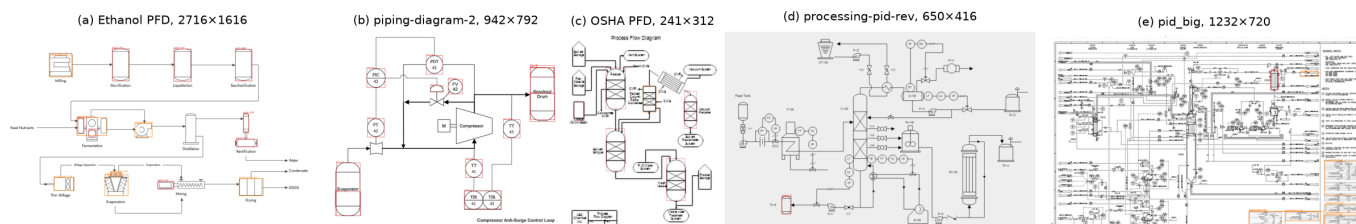


Figure 2: T5+T6 model (seed 0, score ≥ 0.5) on five drawings from outside the dataset. Red = tank, orange = equipment. No ground truth: verdicts are visual judgements.

Drawing (px)	Verdict	What the model did
(a) Ethanol PFD, 2716×1616	Good	Found most major items (3 process tanks, rectification vessels, dryer, evaporator, mill, mixer). Missed the distillation column; labelled the pump as equipment; double-boxed the fermenter.
(b) piping-diagram-2, 942×792	Mixed	Found the evaporator and knockout drum, but called 7 instrument bubbles “tank”. Missed the compressor.
(c) OSHA PFD, 241×312	Fair	Found the reactor, stripper, wash vessel, vacuum receiver, condenser and drum. Missed 3 house-shaped storage tanks and the exchanger.
(d) processing-pid-rev, 650×416	Poor	Found 1 tank of about 11 items. Grey background and low resolution.
(e) pid_big, 1232×720	Poor	Found only condenser 13E02. 6 false alarms on data tables. Missed the column, reactor, vessel, pumps and exchangers.

Findings: the model transfers to clean, reasonably sharp drawings but fails on low-resolution or grey-background images and on symbol styles absent from training. Instrument bubbles and data tables cause false alarms. Instruments belong to the planned small-symbol model, so merging the two models should remove the bubble false alarms. **Input cleanup** (white background, binarize at 2048 px) is not a general fix: (d) improves from 1 to 5 correct items, but the tiny (c) drops from 6 to 0. Across all five, 22 correct before vs 19 after, so it stays opt-in (`--predict-clean`).

Sources

- [1] J. M. Stürmer, M. Graumann, T. Koch. *From Engineering Diagrams to Graphs (PID2Graph)*. arXiv:2411.13929, IEEE DSAA 2025.
- [2] G. Ghiasi et al. *Simple Copy-Paste is a Strong Data Augmentation Method for Instance Segmentation*. CVPR 2021.
- [3] Prasad et al. *SynthPID*. arXiv:2604.16513, CVPR 2026 Workshops.
- [4] F. C. Akyon et al. *Slicing Aided Hyper Inference and Fine-tuning for Small Object Detection (SAHI)*. ICIP 2022.
- [5] M. Gupta, C. Wei, T. Czerniawski. *Semi-supervised symbol detection for piping and instrumentation drawings*. Automation in Construction 159, 2024.
- [6] I. Robinson et al. *RF-DETR*. arXiv:2511.09554, ICLR 2026.
- [7] PyTorch. *torchvision 0.29 Faster R-CNN documentation*; Lightning AI. *torchmetrics 1.9 MeanAveragePrecision documentation*.